

Vector Semantics

Natural Language Processing

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ANN review

- Input
- Feed-forward
- Error
- Gradient
- Using some optimization algorithm we update weights

Word semantics (Lexical Semantics)

- Synonyms: two words are synonymous if they are substitutable one for the other in any sentence without changing the truth conditions of the sentence.
- Antonyms are words with an opposite meaning.

Different view

- Relatedness/association:
 - Coffee and Cup: they are associated in the world by commonly co-participating in a shared event.

Word Semantics (redefined)

- Instead of using some logical language to define each word, we should define words by some representation of how the word was used by actual people in speaking and understanding.
- Define a word by the environment or distribution it occurs in in language use (Joos (1950), Harris (1954), and Firth (1957)).
- A **word's distribution** is the set of contexts in which it occurs, the neighboring words or grammatical environments.
- The idea is that two words that occur in very similar distributions (that occur together with very similar words) are likely to have the same meaning.

Semantic field

- One common kind of relatedness between words is if they belong to the same semantic field semantic field.
- A semantic field is a set of words which cover a particular semantic domain and bear structured relations with each other.

- A word (or sense) is a **hyponym** of another word or sense if the first is more specific, denoting a subclass of the other.
 - For example, car is a hyponym of vehicle.
 - Vehicle is a **hypernym** of car,

Connotation

- The word connotation has different meanings in different fields, but here we use it to mean the aspects of a word's meaning that are related to a writer or reader's emotions, sentiment, opinions, or evaluations.

Concept of distribution

- A word's distribution is the set of contexts in which it occurs, the neighboring words or grammatical environments. The idea is that two words that occur in very similar distributions (that occur together with very similar words) are likely to have the same meaning.

Vector semantics

- The idea of vector semantics is thus to represent a word as a point in some multidimensional semantic space. Vectors for representing words are generally called embeddings , because the word is embedded in a particular vector space.

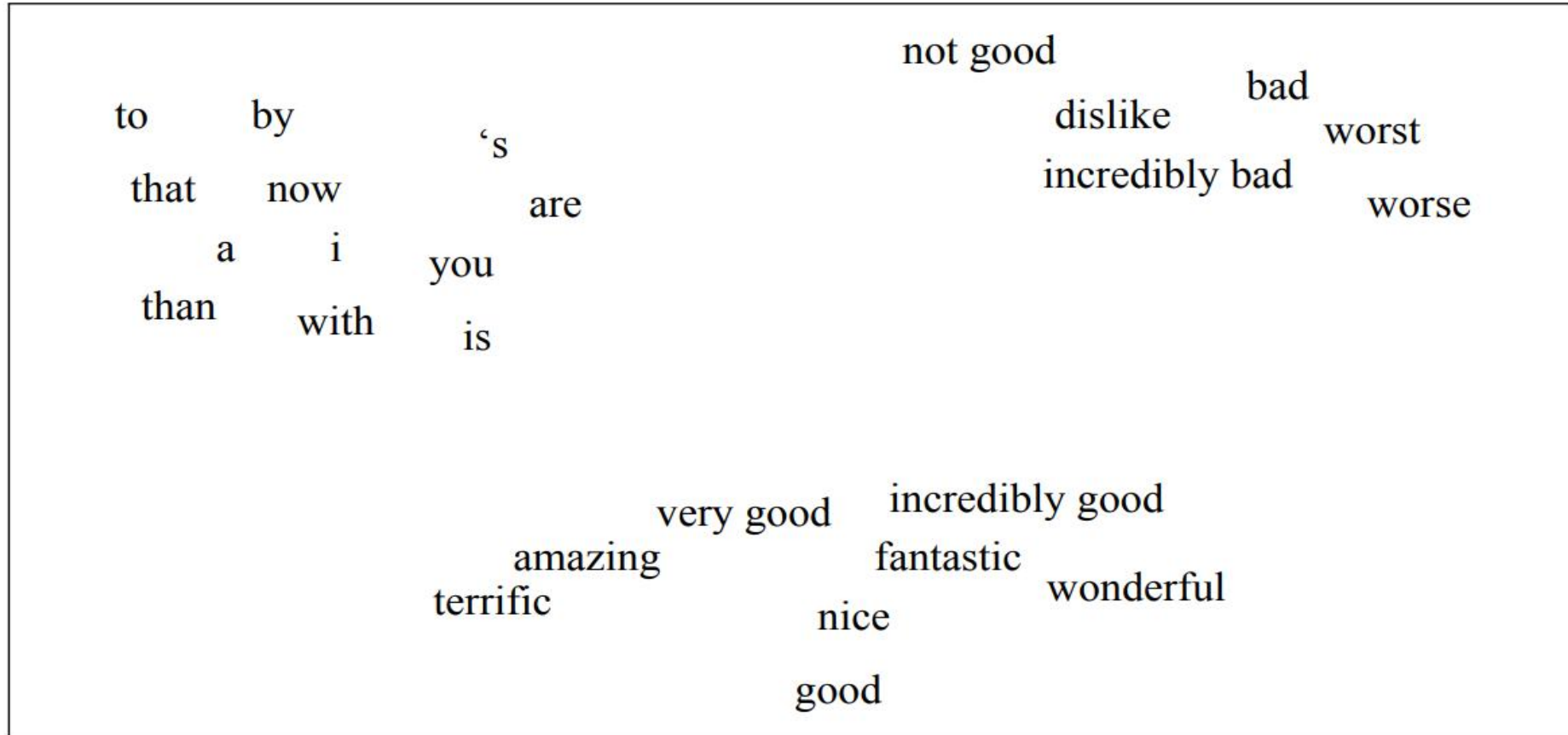


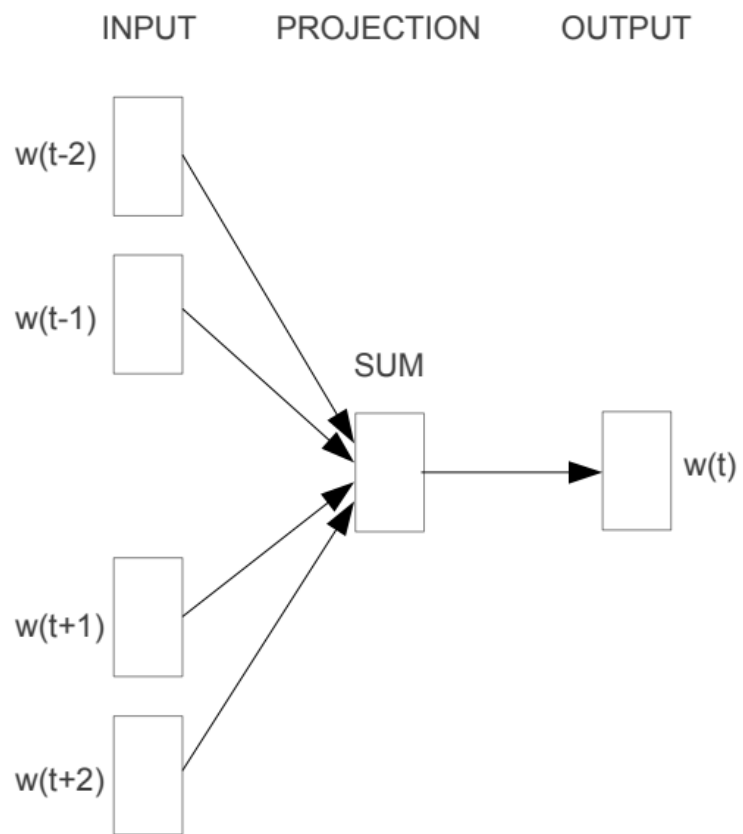
Figure 6.1 A two-dimensional (t-SNE) projection of embeddings for some words and phrases, showing that words with similar meanings are nearby in space. The original 60-dimensional embeddings were trained for a sentiment analysis task. Simplified from [Li et al. \(2015\)](#).

Word2Vec

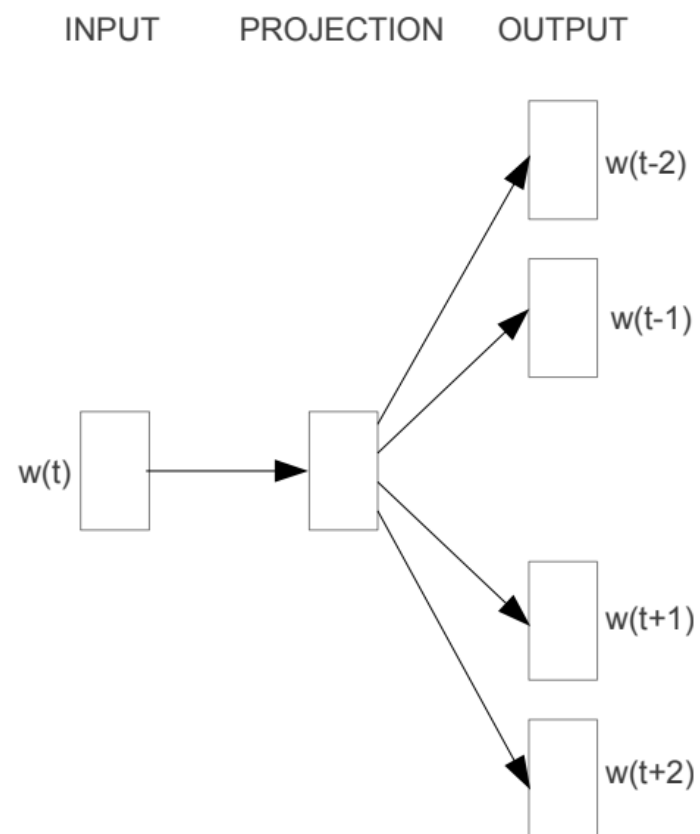
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Word2Vec

- We can train a classifier to predict: “Is word w likely to show up near word w_i or vice versa?”
- We don’t actually care about this prediction task; instead we’ll take the learned classifier weights as the word embeddings.



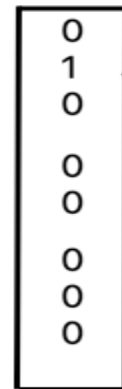
CBOW



Skip-gram

Window size 1
Initial prediction with
random initialization

The
wide
road
shimmered
in
the
hot
sun



$w(t+1)$

0.01
0.01
0.1
0.4
0.04
0.03
0.08
0.02

The
wide
road
shimmered
in
the
hot
sun

$w(t-1)$

0.01
0.01
0.5
0.2
0.04
0.03
0.08
0.02

The
wide
road
shimmered
in
the
hot
sun

Back propagation

y_{pred}

0
0
0
1
0
0
0
0

y

0
0
1
0
0
0
0
0

Loss

0
0
1
0
0
0
0
0

1
0
0
0
0
0
0
0

Window size 1

Prediction after backpropagation

The
wide
road
shimmered
in
the
hot
sun

0
1
0
0
0
0
0
0



$w(t+1)$

0.01
0.01
0.8
0.1
0.04
0.03
0.08
0.02

The
wide
road
shimmered
in
the
hot
sun

$w(t-1)$

0.79
0.01
0.1
0.09
0.04
0.03
0.08
0.02

The
wide
road
shimmered
in
the
hot
sun

Back propagation

y_{pred}

0
0
1
0
0
0
0
0

y

0
0
1
0
0
0
0
0

Loss

